

LATENT PLANNING VIA EMBEDDING ARITHMETIC: A CONTRASTIVE APPROACH TO STRATEGIC REASONING

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ABSTRACT

Planning in high-dimensional decision spaces increasingly depends on learned representations, yet most latent planning systems still rely on learned dynamics, policies, or value functions. This paper establishes that, when reliable state evaluations are available, planning can instead be performed directly in an evaluation-aligned embedding space. We introduce SOLIS, a supervised-contrastive framework that organizes positions so that proximity reflects outcome similarity and a single global *advantage vector* orients the representation from losing to winning regions. Action selection then reduces to ranking legal continuations by their alignment with this direction, replacing policy inference or value-head evaluation with vector operations in latent space. In chess, SOLIS attains competitive strength with shallow embedding-guided search under constrained conditions. These results identify evaluation-aligned latent planning as a lightweight and interpretable alternative to conventional dynamics models, policy learning, and broad rollout-based search within the demonstrated setting.

1 INTRODUCTION

Planning in high-dimensional state spaces remains a central problem for artificial intelligence because action sequences expand combinatorially with decision horizon. Conventional reinforcement learning and game-playing systems address this difficulty through explicit search, learned policies, or value estimation, but these mechanisms can become computationally demanding as the action space grows (Jiang et al., 2023; Gieslmann and Pokorny, 2022). Latent planning offers a complementary strategy: it replaces reasoning over raw states with reasoning over compact learned representations, thereby reducing search complexity while preserving task-relevant structure (Hafner et al., 2019; 2020; Schrittwieser et al., 2020).

We test a stronger form of latent planning: an embedding space that supports action selection without an explicit policy head, value head, or learned dynamics model. We introduce **SOLIS** (*Strategic Optimization via Latent Interpolative Spaces*), a chess engine whose transformer encoder is trained with supervised contrastive learning to map board states into an evaluation-aligned latent space. The representation is constructed so that distance encodes similarity of outcome, while a single global “advantage vector” provides an operational direction from losing to winning regions. SOLIS exploits this geometry by ranking candidate moves according to their alignment with that vector, making planning a procedure over latent-space arithmetic rather than over learned action probabilities.

The empirical results show that this representation is not merely descriptive but operationally useful for search. With a 5-ply search under 50 ms/move blitz conditions, SOLIS reaches an Elo of 2500+; in equal-depth matches at search depth $S \in \{3, 4, 5\}$ with a fixed branching factor, it is competitive with the configured Stockfish baseline used in our experiments. Game-level visualizations further show that decisive trajectories move coherently through the embedding space, providing qualitative evidence that the learned geometry exposes interpretable structure relevant to planning.

This work contributes a representation-first formulation of latent planning. In contrast to methods that couple latent dynamics with policy or value heads (Hafner et al., 2019; 2020; Schrittwieser et al., 2020), SOLIS uses the geometry of an evaluation-aligned space itself as the scoring mechanism for action selection. Chess provides a demanding perfect-information testbed, and the same formulation

is applicable, within the limits of our evidence, to domains in which state evaluations can be obtained or learned and legal continuations can be enumerated.

2 BACKGROUND

Planning in large action spaces is difficult because every additional decision step expands the set of possible futures exponentially. The methods most relevant to SOLIS therefore differ in how they represent states, how they obtain evaluative information, and how they allocate computation between local search and globally learned solutions. This section situates our approach within Markov decision processes, model-based reinforcement learning, chess engines, and contrastive representation learning.

2.1 MARKOV DECISION PROCESSES

Sequential decision-making problems are commonly formalized as Markov Decision Processes (MDPs), specified by the tuple $\{\mathcal{S}, \mathcal{A}, T, R, p(s_0), \gamma\}$. Here $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, T is the transition model, R is the reward function, $p(s_0)$ is the initial-state distribution, and $\gamma \in [0, 1]$ discounts future rewards (Puterman, 2014). A policy $\pi : \mathcal{S} \rightarrow p(\mathcal{A})$ maps each state to a probability distribution over actions.

Within this formalism, a *model* provides access to the MDP dynamics: given a state–action pair, it returns or predicts the successor state and reward (Moerland et al., 2022). In rule-governed games such as chess and Go, the environment supplies this model exactly. In domains without known rules, the transition and reward structure must instead be approximated from data (Gieselmann and Pokorný, 2022).

Planning and reinforcement learning are distinguished by the way they use this dynamical access (Moerland et al., 2022). Planning methods assume a model and simulate forward from the current state in order to select an action, analogous to deliberate mental search (Russell and Norvig, 1995). Because the resulting computation is conditioned on the current state, it is *local* and can be discarded after execution. Reinforcement learning methods, by contrast, typically operate without reversible access to the MDP dynamics (Sutton and Barto, 1998). Through sequential interaction, they learn a *global solution*—such as a policy or value function—that generalizes across the state space.

Model-based reinforcement learning combines these perspectives by using a model while learning a global solution. As summarized by Moerland et al. (2022), this combination appears in two principal forms. In model-based RL with a learned model, exemplified by Dyna (Sutton, 1991), both the dynamics and the policy or value representation are learned. In model-based RL with a known model, exemplified by AlphaZero (Silver et al., 2017), the transition dynamics are supplied and learning focuses on the global policy and value components.

2.1.1 MODEL-BASED RL WITH A LEARNED MODEL

A substantial model-based RL literature learns both latent dynamics and a global decision mechanism. PlaNet (Hafner et al., 2019) learns a latent dynamics model and couples it to a learned global solution. Dreamer (Hafner et al., 2020) and its successors DreamerV2 (Hafner et al., 2022) and DreamerV3 (Hafner et al., 2024) extend this paradigm to more complex tasks while retaining learned value functions and policies. MuZero (Schrittwieser et al., 2020) similarly learns latent dynamics, but strengthens action selection by integrating the model with tree search while also training global policy and value functions. Dyna (Sutton, 1991) and MBPO (Janner et al., 2021) provide additional examples in which learned models support value or policy learning.

2.1.2 MODEL-BASED RL WITH A KNOWN MODEL

When the environment dynamics are available, algorithmic design can concentrate on how search and learning should be coupled. AlphaZero (Silver et al., 2017) uses exact game rules as its model and combines Monte Carlo tree search (MCTS) with learned policy and value heads. SAVE (Hamrick et al., 2020) makes a related simulator assumption and integrates Q-learning with MCTS: a

learned Q-function initializes or guides search, and Q-estimates produced by search are amortized back into the network.

2.1.3 LATENT PLANNING

Latent planning reduces decision complexity by carrying out search in a compressed representation rather than in the raw state space. PlaNet (Hafner et al., 2019), Dreamer (Hafner et al., 2020), and MuZero (Schrittwieser et al., 2020) combine latent dynamics with policy or value heads, enabling simulated rollouts in learned coordinates. Expansive Latent Space Trees (ELAST) (Gieselmann and Pokorný, 2022) formulates latent planning as exploratory tree search, using contrastive representations and sampling-based expansion for long-horizon control. Trajectory Autoencoding Planner (TAP) (Jiang et al., 2023) also compresses the planning problem, learning a latent action space and searching over discrete action codes.

SOLIS occupies a different point in this design space. It does not train a policy head, a value head, or a learned dynamics model; instead, it distills evaluation targets into the geometry of the representation itself. At inference time, planning proceeds by selecting continuations whose embeddings move toward regions associated with favorable outcomes.

2.2 CHESS ENGINES

Computer chess has long served as a demanding benchmark for search and evaluation. IBM’s Deep Blue demonstrated the power of scaled chess computation in 1997 by defeating reigning world champion Garry Kasparov. Early versions of Stockfish (Romstad et al., 2008) followed the same broad template: handcrafted evaluation functions coupled to deep alpha-beta search. This paradigm achieved superhuman play, but its strength remained closely tied to the quality and coverage of human-designed static evaluation features.

The limitations of static evaluation motivated engines that reduce dependence on handcrafted chess knowledge. AlphaZero (Silver et al., 2017) and its open-source counterpart Leela Chess Zero (Pascutto and Linscott, 2018) learned position evaluation through self-play reinforcement learning, using no prior chess knowledge beyond the rules. Modern Stockfish incorporated an efficiently updatable neural-network evaluation module (NNUE) (Nasu, 2018), improving evaluation while preserving the alpha-beta search infrastructure.

Although these systems now far exceed human strength, much of their search structure still reflects principles introduced by v. Neumann (1928) and Shannon (1950) and later embodied in Deep Blue: deep alpha-beta exploration over very large numbers of states (Lai, 2015). SOLIS addresses this persistent reliance on broad expansion by asking whether contrastive learning can make the representation itself evaluative enough to support low-branching search through a high-dimensional embedding space.

2.3 CONTRASTIVE LEARNING

Contrastive representation learning trains an encoder to place similar inputs near one another and dissimilar inputs farther apart in latent space (Chen et al., 2020). Given a batch I containing an anchor x_i , the encoder $f(\cdot)$ maps the input to a normalized embedding $z_i = f(x_i)$. Positive pairs (z_i, z_j) encode the relevant notion of semantic similarity, whereas the remaining in-batch samples function as negatives.

A common contrastive objective is InfoNCE (van den Oord et al., 2019), defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

where $A(i) = I \setminus \{i\}$ contains all batch elements except anchor i , $\text{sim}(\cdot, \cdot)$ denotes the similarity function, and τ is a scalar temperature. In self-supervised contrastive learning, z_j is typically an augmentation of the anchor z_i , so no external labels are required (Chen et al., 2020; Khosla et al., 2021).

When labels are available, the supervised contrastive loss (SupCon) (Khosla et al., 2021) extends InfoNCE by allowing multiple positive examples per anchor. The SupCon loss is defined as:

$$\mathcal{L}_{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)}, \quad (2)$$

where $P(i)$ is the set of positives associated with anchor i , and $A(i)$ is defined as above. By permitting multiple positives per anchor, SupCon can exploit richer label structure, and larger sets of positive and negative pairs have been shown to improve representation quality (Khosla et al., 2021; Chowdhury et al., 2024).

3 METHODS

We next specify the SOLIS pipeline: FEN tokenization, transformer encoding, supervised contrastive training, and the embedding-guided inference procedure that converts the learned geometry into action scores.

3.1 INPUT REPRESENTATION AND TOKENIZATION

Each chess state is represented by its Forsyth-Edwards Notation (FEN) ASCII string, which compactly specifies piece placement, side to move, castling rights, en passant target, half-move clock, and full-move counter. Following the tokenization scheme of Ruoss et al. (2024), we expand run-length encodings and convert every FEN into a fixed-length sequence of 77 tokens. This representation omits historical information that can affect certain chess rules, including threefold repetition and the fifty-move rule; as noted by Ruoss et al. (2024), the resulting formulation is therefore non-Markovian in these special cases. We return to this limitation in Section 5.

3.2 ENCODER ARCHITECTURE

The SOLIS encoder is a multi-layer transformer (Vaswani et al., 2023) adapted to fixed-length tokenized chess positions. We evaluate two capacity regimes, SOLIS Base and SOLIS Mini, whose architectural configurations are reported in Table 1.

Table 1: Transformer encoder configurations evaluated in the SOLIS experiments.

Model	Layers	Embedding dim. D	Heads	MLP Size	Params
SOLIS Mini	6	128	8	256	0.8M
SOLIS Base	6	1024	16	1024	41M

A learned token embedding matrix maps each input token to a D -dimensional vector. We prepend a classification token (CLS) (Devlin et al., 2019; Dosovitskiy et al., 2021) to aggregate sequence-level information, add learned positional encodings to the fixed-length token sequence (Ruoss et al., 2024), and process the result with stacked transformer encoder layers using GELU (Hendrycks and Gimpel, 2023) activations and dropout (Srivastava et al., 2014). The final hidden state of the CLS token is then projected linearly and ℓ_2 normalized to produce the embedding used for contrastive training and search.

3.3 SUPERVISED CONTRASTIVE TRAINING

We train the encoder on 5 million randomly sampled ChessBench positions (Ruoss et al., 2024), each annotated with a Stockfish-estimated win probability for the player to move. All evaluations are normalized to White’s perspective, so values near 1.0 denote a decisive White advantage and values near 0.0 denote a decisive Black advantage.

Positive pairs are defined by evaluation similarity. We set δ as the margin for similar positions; for each anchor, we precompute all positions whose win probabilities differ by less than δ and randomly sample five positives during training. The batch mask is constructed by comparing all in-batch win

probabilities and marking a pair as positive when its evaluation difference is below δ . This margin-based criterion draws together positions with nearly equal evaluations while separating positions with larger evaluation gaps, thereby imposing an ordered latent geometry from losing to winning regions.

We optimize the encoder with supervised contrastive learning (SupCon) (Khosla et al., 2021) using these masks to specify positive and negative pairs. Pairwise scores use cosine similarity, and the temperature $\tau = 0.07$ (Girdhar et al., 2023; Radford et al., 2021) controls the concentration of the contrastive distribution. Optimization uses stochastic gradient descent with momentum 0.9 (Dosovitskiy et al., 2021; Chen et al., 2020; He et al., 2015), batch size 128, and 400,000 training steps. Training is performed with data parallelism (Paszke et al., 2019) on six NVIDIA L40S GPUs with 48 GB of memory each. Appendix A.2 reports the complete hyperparameter configuration.

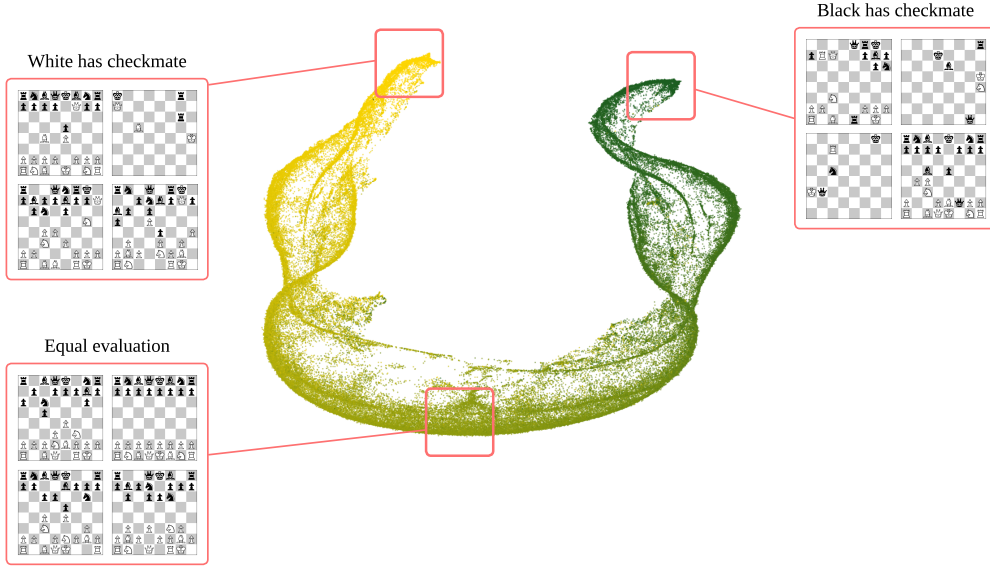


Figure 1: UMAP projection of the Base-model embedding space, with each position colored by Stockfish-evaluated win probability (gold = White favored, green = Black favored).

Figure 1 visualizes the Base-model representation using a two-dimensional UMAP projection (McInnes et al., 2020). Each point is an encoded test position, colored by its Stockfish-evaluated win probability for White. The projection displays a coherent evaluation gradient: positions with strong White advantage appear in gold (*White has checkmate*), whereas positions with strong Black advantage appear in dark green (*Black has checkmate*).

3.4 EMBEDDING-GUIDED SEARCH

At inference, SOLIS plans directly in the learned representation by assigning scalar scores to legal continuations with a single global advantage direction. We estimate the endpoint means from held-out positions at the extremes of the evaluation scale,

$$\mu_{\text{White}} = \mathbb{E}[z \mid p=1.0], \quad \mu_{\text{Black}} = \mathbb{E}[z \mid p=0.0],$$

and define the normalized advantage vector

$$\vec{a} = \frac{\mu_{\text{White}} - \mu_{\text{Black}}}{\|\mu_{\text{White}} - \mu_{\text{Black}}\|}.$$

We evaluate two scoring mechanisms that convert this direction into scalar move preferences.

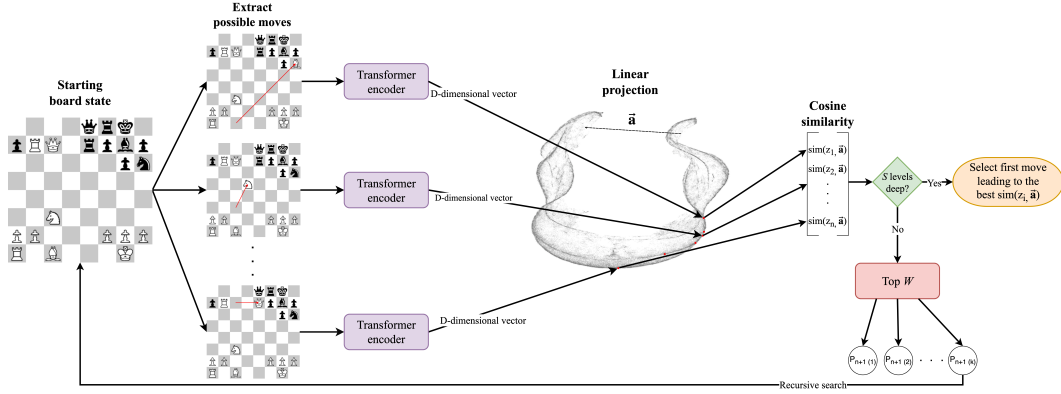


Figure 2: System overview of SOLIS. Legal child positions are embedded by the transformer encoder, scored by alignment with the global advantage vector, and pruned to the top- W continuations. For simplicity, the diagram depicts the unanchored score $\cos(z', \vec{a})$.

3.4.1 UNANCHORED (GLOBAL-DIRECTIONAL)

The unanchored mechanism scores each child embedding by its direct alignment with the global advantage direction. For a child embedding z' , the score is

$$\text{score}_{\text{dir}}(z') = \langle z', \vec{a} \rangle,$$

so higher-ranked moves are those whose embeddings project farthest along the advantage direction, independent of any explicit origin. Figure 2 illustrates this scoring mechanism.

3.4.2 ANCHORED (RELATIVE PROJECTION)

The anchored mechanism first translates each child embedding relative to the Black endpoint mean and then scores

$$\text{score}_{\text{anch}}(z') = \langle z' - \mu_{\text{Black}}, \vec{a} \rangle,$$

so that the scalar value is the signed displacement from μ_{Black} along $\vec{a}$. This anchoring removes dependence on an arbitrary coordinate origin and, in our experiments, provides a more stable basis for comparing continuations.

Both scoring mechanisms use the same search procedure, summarized in Figure 2. From the “Starting board state”, SOLIS “Extract possible moves”, embeds each child position in parallel with the “Transformer encoder”, computes either $\text{score}_{\text{dir}}$ or $\text{score}_{\text{anch}}$ in the “Cosine similarity” block, and retains the “Top- W ” children. If the search has reached “ S levels deep?”, the algorithm terminates and “Select first move. . .”; otherwise, it continues the “Recursive search” under a min–max preference rule in which White maximizes and Black minimizes. We use Zobrist hashing (Zobrist, 1990) to implement a transposition table and avoid redundant evaluation of repeated positions.

4 RESULTS

4.1 RATING CALCULATION

We estimate Elo by evaluating each SOLIS configuration against Stockfish under a fixed 50 ms per-move time limit (Ruoss et al., 2024); Appendix A.3 specifies the Stockfish configuration. The evaluation varies both search depth and search width, and each depth/width configuration plays more than 600 games with alternating colors against Stockfish instances configured at multiple Elo caps. Both unanchored and anchored planning are evaluated under this protocol. The caps are selected so that each SOLIS configuration includes at least one matchup with a positive win rate and one with a negative win rate, enabling stable rating estimation. Elo ratings and 95% confidence intervals are computed with Bayesian logistic regression using BayesElo (Coulom, 2008) and its default confidence parameter of 0.5. Figure 3 reports the unanchored ratings, Figure 4 reports the anchored ratings, and Appendix A.6 gives the Stockfish caps and complete match outcomes used for reproducibility.

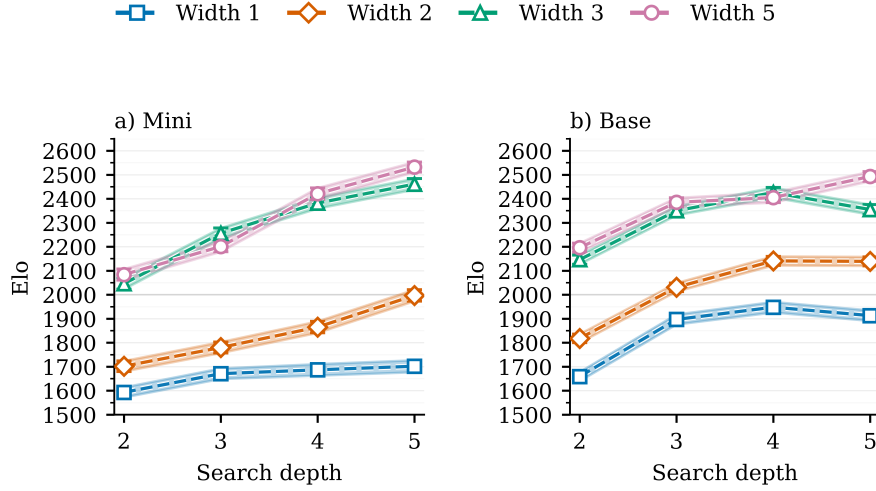


Figure 3: Estimated Elo for SOLIS Mini and SOLIS Base under unanchored planning across search widths and depths. Shaded bands denote 95% confidence intervals.

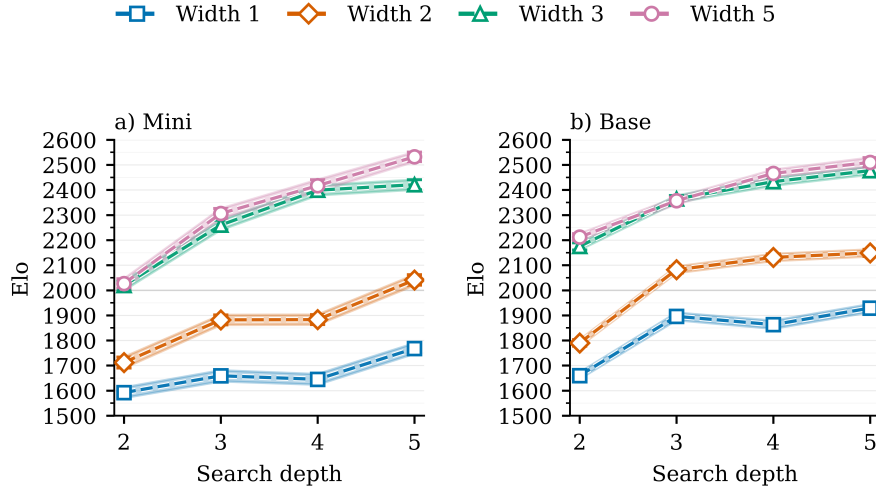


Figure 4: Estimated Elo for SOLIS Mini and SOLIS Base under anchored planning across search widths and depths. Shaded bands denote 95% confidence intervals.

Unanchored ratings Under unanchored planning, Elo increases systematically as search becomes deeper. Depth accounts for the largest gains, whereas width produces a pronounced improvement from $W \in \{1, 2\}$ to $W=3$ and little additional benefit at $W=5$. This pattern indicates that, for this scoring rule, the most useful continuations are usually contained among the top three embedded candidates. SOLIS Base leads Mini by approximately 100–250 Elo at shallow depths; at depths 4 and 5, Mini becomes competitive with Base and exceeds it for $W \in \{3, 5\}$ under this planning mechanism.

Anchored ratings Anchored scoring produces a smoother scaling profile across both depth and width. The Base–Mini gap remains approximately 100–250 Elo, and both models exceed 2500 Elo at depth 5. Unlike the unanchored setting, SOLIS Base is consistently stronger than Mini across the evaluated configurations. Anchoring also yields a steady 10–30 Elo improvement for Base, supporting the interpretation that measuring displacement relative to the Black mean provides a more reliable basis for comparing continuations.

4.2 EQUAL-DEPTH SIMULATIONS

We next compare SOLIS and Stockfish in equal-depth head-to-head simulations. Guided by the Elo ablation in Section 4.1, SOLIS uses a fixed branching factor of $W=3$, since wider searches produce only marginal additional gains and the strongest continuations are generally among the top three candidates. Stockfish uses its default expansion procedure and is not constrained to a fixed branching factor. For each depth $S \in \{3, 4, 5\}$, we simulate 200 games with alternating colors. Table 2 reports the match points for both unanchored and anchored scoring.

Simulation results With unanchored scoring, SOLIS Base obtains more match points than Stockfish at every tested depth. SOLIS Mini is essentially competitive at depths 3 and 5 but falls behind at depth 4. With anchored scoring, Base loses narrowly at depth 3, then surpasses Stockfish at depths 4 and 5; Mini wins the depth-3 and depth-5 comparisons but again trails at depth 4. These results show that the embedding-guided search remains competitive against the configured Stockfish baseline even when SOLIS expands only a small, fixed number of continuations.

Table 2: Equal-depth match points for SOLIS versus Stockfish across depths $S \in \{3, 4, 5\}$. The middle columns report unanchored direct-similarity planning, and the right columns report anchored projection. Higher scores are better; draws count as 0.5.

Config	Depth	Direct Similarity		Anchored Projection	
		SOLIS score	Stockfish score	SOLIS score	Stockfish score
Base	3	106	94	98	102
	4	109	91	108	92
	5	107.5	92.5	110	90
Mini	3	108	92	105	95
	4	85	115	91	109
	5	101	99	113	87

Nodes searched With fixed branching factor $W=3$, SOLIS searches at most $N(S) = \sum_{i=0}^S 3^i$, nodes at depth S , giving $N(3) = 40$, $N(4) = 121$, and $N(5) = 364$. Stockfish searches a variable number of nodes per move because its expansion depends on pruning and internal heuristics: at depth 3 it searched 137 nodes on average (maximum 1625), at depth 4 it averaged 214 nodes (maximum 4463), and at depth 5 it averaged 338 nodes (maximum 10611). SOLIS remains competitive with this Stockfish configuration while using a fixed and substantially smaller search envelope, indicating that **much of the strength normally gained from broad expansion can be recovered with higher-quality representations**.

4.3 INTERPRETABLE GAME TRAJECTORIES

To assess the interpretability of the learned geometry, we visualize move sequences from real games in the shared embedding space. Figure 5 presents three representative trajectories: a White win, a balanced draw, and a Black win. Each position is embedded independently, projected into the same two-dimensional coordinate system, and connected with arrows to show temporal progression. Decisive games tend to trace smooth paths toward the corresponding winning region, whereas balanced games remain near the center of the projection. These trajectories complement Figure 1 by showing that the evaluation-aligned structure persists along actual game sequences, not only in static position samples.

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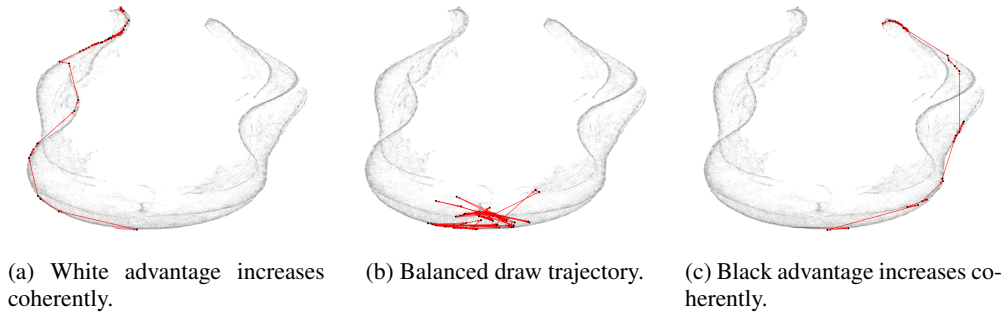


Figure 5: Latent trajectories for three games embedded in the shared SOLIS representation. Red arrows indicate the chronological progression of positions through the evaluation-aligned space.

5 DISCUSSION

5.1 LIMITATIONS

Blindness to threefold repetition SOLIS selects moves from the current position by measuring alignment with the advantage vector, which keeps inference lightweight but omits move history. This omission matters for rules whose legality or strategic value depends on previous states. The clearest case is threefold repetition: if the same position, including side to move, castling rights, and en-passant rights, occurs three times, either player may claim a draw. An opponent can therefore use repeated positions, including perpetual-check patterns, to neutralize positions that SOLIS would otherwise be expected to convert. Game C in Appendix A.4 gives an example in which SOLIS reaches a winning position against Stockfish but allows a premature draw by repetition.

Indecisiveness in winning positions Because SOLIS is trained on win-probability targets, it does not distinguish forcing lines by their length once they share the same terminal evaluation. A mate-in-1 and a mate-in-20 can both have win probability 1.0, even though the shorter continuation is strategically preferable. Selecting the longer line can give the opponent additional opportunities to complicate the position, sometimes producing unnecessary draws or losses. Ruoss et al. (2024) observed the same issue and avoided it by consulting Stockfish as an external oracle. SOLIS instead relies on its own latent scoring procedure, leaving forced-win prioritization as an important remaining limitation. Game I in Appendix A.5 illustrates a case in which SOLIS has many winning continuations for several consecutive moves but chooses suboptimal lines that lead to a draw.

5.2 CONCLUSION

This paper demonstrates that planning can be performed directly in an evaluation-aligned latent space without training a policy head, value head, or learned dynamics model. SOLIS learns a supervised-contrastive embedding in which proximity reflects outcome similarity and a single global *advantage vector* orients the space from losing to winning regions. Action selection is then implemented as vector arithmetic: legal continuations are embedded, ranked by alignment with this direction, and expanded through a small fixed-width search.

In chess, SOLIS uses a fixed branching factor $W=3$ and shallow depths $S \in \{3, 4, 5\}$. Under these conditions, it matches or exceeds the time-capped Stockfish configuration in several equal-depth matchups while searching far fewer nodes per move: 40–364 nodes for SOLIS, compared with hundreds to thousands for Stockfish at the same depths. Elo increases with depth; with anchored scoring, both Mini and Base exceed 2500 Elo at depth five, and anchoring provides a consistent 10–30 Elo gain for Base. Search width contributes little beyond three candidates, supporting the conclusion that the learned representation, rather than broad expansion alone, supplies much of the playing strength. The same embedding also produces interpretable trajectories: decisive games move toward winning regions, while drawn games remain near neutral areas.

Together, the results support evaluation-aligned latent planning as a simple, compute-efficient alternative to broad rollout-based search and explicit dynamics modeling in the tested perfect-

information setting. The formulation is naturally relevant to other domains in which state evaluations are available or learnable and low-branching search is valuable, although transfer beyond chess remains to be tested. Future work should add rule-aware state features, prefer shorter forced wins, combine SOLIS with policy priors or MCTS, study calibration and uncertainty, and evaluate transfer beyond chess.

REPRODUCIBILITY STATEMENT

Appendix A.2 reports the training setup, hyperparameters, and implementation details required to reproduce the learned representations. Section 3 specifies the encoder architecture, input representation, supervised contrastive objective, and inference procedure, with model configurations summarized in Table 1. Section 4 and Appendix A.3 define the evaluation protocol, including match setup, Stockfish configuration, and rating estimation. Appendix A.6 provides the detailed match outcomes used to verify the Elo estimates.

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A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models were used only for minor manuscript-support tasks, including formatting and figure/table placement. They were not used to design experiments, analyze results, or generate technical content. All methodological and experimental contributions were developed and verified by the authors.

A.2 TRAINING SETUP

Both SOLIS configurations, Mini and Base, are trained with supervised contrastive learning on 5M ChessBench positions. Training uses cosine similarity over ℓ_2 -normalized embeddings and SGD with momentum on 6 48 GB L40S GPUs for 400k steps with batch size 128.

Among these settings, τ and δ most directly determine the geometry of the contrastive space. The temperature $\tau = 0.07$ follows prior contrastive work (Radford et al., 2021; Girdhar et al., 2023): smaller values sharpen similarity distributions, whereas larger values smooth gradients excessively. We trained with $\tau \in \{0.05, 0.07, 0.1\}$ and found that 0.07 converged most reliably across model sizes. The margin $\delta = 0.05$ balances two failure modes—collapsing too many positions into a single neighborhood when δ is large and fragmenting evaluation-equivalent regions when δ is small—and is consistent with Stockfish evaluation noise, since differences below 0.05 are rarely meaningful in practice.

A.3 STOCKFISH SETUP

All experiments use the official Linux release of Stockfish 16 (Romstad et al., 2008) with 32 MB hash size and default threading. Rating-capped opponents are configured through `UCI.LimitStrength` and `UCI.Elo`, and fixed-depth opponents are configured with the `depth` parameter. For node-count measurements, the engine information level is set to `INFO.ALL`, and the reported `nodes` attribute is extracted for each move.

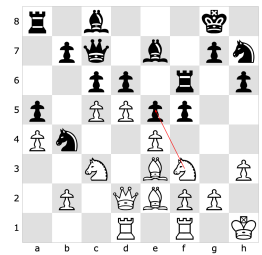
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Table 3: Training hyperparameters used for both SOLIS Mini and SOLIS Base.

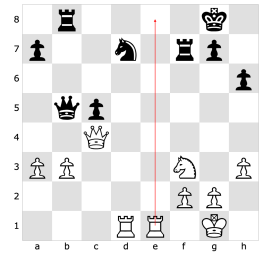
Attribute	Value
Loss	SupCon
τ	0.07
δ (positive margin)	0.05
Positives per anchor	5
Optimizer	SGD
Momentum	0.9
Learning rate	0.05
Batch size	128
Steps	400k
Dropout	0.10
Activation	GELU
Similarity	cosine
Embedding norm	ℓ_2
Token length	77

A.4 REPRESENTATIVE GAMES: STOCKFISH (CAP) VS. SOLIS (DEPTH, WIDTH)

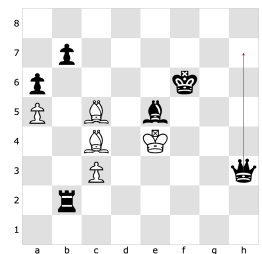
The following games illustrate how SOLIS behaves against rating-capped Stockfish when SOLIS is constrained to search depth S and width W while Stockfish searches without a fixed width. Games *A*, *B*, and *E* show tactical resourcefulness and longer-horizon planning within the embedding-guided search. Games *C* and *D* highlight representative failure modes. In each table, the left column gives the complete game in Portable Game Notation (PGN), and the right column shows a critical position.

<pre> 1. d4 Nf6 2. c4 d6 3. Nc3 e5 4. Nf3 Qe7 5. e4 c6 6. d5 Qc7 7. Be2 h6 8. a4 a5 9. O-O Be7 10. h3 Na6 11. Be3 Nb4 12. Kh1 Nh7 13. Qd2 O-O 14. Rad1 f5 15. c5 Rf6 16. Nxe5 f4 17. Bxf4 Rxf4 18. Qxf4 dxe5 19. d6 exf4 20. Bc4+ Kf8 21. dxc7 Ke8 22. e5 Ng5 23. Rd6 Bxd6 24. exd6 Bd7 25. Re1+ Kf8 26. Re7 Bf5 27. h4 Nd5 28. Nxd5 cxd5 29. Bxd5 Be6 30. Bxe6 f3 31. hxg5 b6 32. Bh3 hxg5 33. Re5 Rc8 34. Rf5+ Kg8 35. Rf7 g4 36. cxb6 g6 37. d7 fxg2+ 38. Kh2 Rf8 39. Bxg2 Kxf7 40. Bc6 Ke7 41. Kg3 Rf3+ 42. Kh2 Rd3 43. b7 Rh3+ 44. Kg2 Re3 45. d8=Q+ Kf7 46. b8=Q Rg3+ 47. Kh2 Rh3+ 48. Kg1 Rh7 49. Qe8+ Kf6 50. Qbd8+ Kf5 51. Be4+ Kf4 52. Qf6# 1-0 </pre>	
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Game A — SOLIS ($S=5$, $W=3$) vs. SF 2350, SOLIS–White. Result: 1–0. SOLIS converts a knight sacrifice into a concrete middlegame initiative and decisive attack.

<pre> 1. d4 d5 2. e3 c6 3. Bd3 h6 4. h3 Nf6 5. Nf3 e6 6. O-O b6 7. c4 Ba6 8. b3 Nbd7 9. Bb2 Bb4 10. a3 Be7 11. Nbd2 O-O 12. Re1 c5 13. Qe2 Qc7 14. e4 Nxe4 15. Nxe4 dxe4 16. Bxe4 Rab8 17. d5 Bf6 18. dxe6 Bxb2 19. exf7+ Rxf7 20. Qxb2 Bb7 21. Bxb7 Qxb7 22. Rad1 b5 23. Qe2 bxc4 24. Qxc4 Qb5 25. Re8+ Nf8 26. Rxb8 Qxb8 27. Rd5 Qf4 28. Qxf4 Rxf4 29. Rxc5 Ne6 30. Rc6 Nd4 31. Nxd4 a5 32. Rc4 Rf8 33. Nc6 Kh8 34. Nxa5 Ra8 35. b4 Rd8 36. g3 Rd3 37. a4 g5 38. b5 Rd2 39. b6 Rd5 40. b7 Rxa5 41. Rc8+ Kh7 42. b8=Q Rd5 43. Rc7+ Kg6 44. Qg8+ Kf5 45. Qxd5+ Kg6 46. Qf7# 1-0 </pre>	
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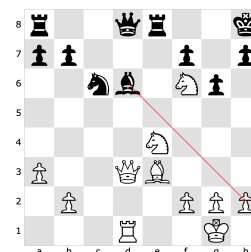
Game B — SOLIS ($S=5$, $W=3$) vs. SF 2300, SOLIS–White. Result: 1–0. SOLIS sacrifices a rook to win Stockfish’s queen and converts the resulting material advantage.

<pre> 1. e3 Nf6 2. Nc3 d5 3. Nf3 c5 4. d4 e6 5. g3 a6 6. a4 Nc6 7. Ne2 Bd6 8. Bg2 O-O 9. h3 Re8 10. c3 Qc7 11. Nd2 e5 12. Nb3 c4 13. dxe5 Nxe5 14. Nbd4 Nd3+ 15. Kf1 Nc5 16. Bf3 Bd7 17. Kg2 Be5 18. a5 Rad8 19. Qc2 g6 20. Bd2 h5 21. Be1 h4 22. gxh4 Kg7 23. Ng3 Rh8 24. h5 Rdg8 25. hxg6 fxg6 26. Be2 Qd6 27. Qd1 Qe7 28. f3 Bd6 29. Nf1 Rh5 30. h4 Rgh8 31. Bf2 R5h7 32. Qe1 Bb8 33. Bg3 Ba7 34. Bf2 Nh5 35. Ng3 Bb8 36. f4 Nf6 37. h5 Rh6 38. hxg6 Rxh1 39. Qxh1 Qe8 40. Ndf5+ Bxf5 41. Nxf5+ Kxg6 42. Ne7+ Kf7 43. e4 Rxh1 44. Rxh1 Qxe7 45. Bxc5 Qxe4+ 46. Bf3 Qg6+ 47. Kf1 Bxf4 48. b4 cxb3 49. Rh8 Ke6 50. Rf8 b2 51. Rxf6+ Kxf6 52. Bxd5 b1=R+ 53. Ke2 Rb2+ 54. Kf3 Be5 55. Bc4 Qh5+ 56. Ke3 Qh3+ 57. Ke4 Qh7+ 58. Kf3 Qh3+ 59. Ke4 Qh7+ 60. Kf3 Qh3+ 61. Ke4 Qh7+ 62. Kf3 Qh3+ 63. Ke4 Qh7+ 64. Kf3 Qh3+ 65. Ke4 1/2-1/2 </pre>	
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Game C — SF 2200 vs. SOLIS ($S=5$, $W=3$), SOLIS–Black. Result: 1/2–1/2. SOLIS emerges with a winning endgame but, because the representation omits repetition history, permits a threefold-repetition draw.

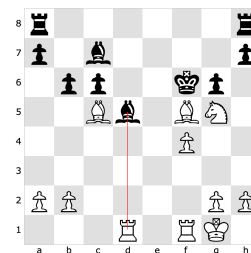
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1. d4 d5 2. c4 dxc4 3. e3 e5 4. Bxc4 exd4 5. exd4
Nf6 6. Nc3 Bd6 7. Qe2+ Qe7 8. Be3 O-O 9. Nf3 c6 10.
O-O Be6 11. Rfd1 Re8 12. d5 cxd5 13. Bxd5 Nc6 14.
a3 Nxd5 15. Nxd5 Qd8 16. Qd3 Bg4 17. Ng5 g6 18. Ne4
Bxd1 19. Ndf6+ Kh8 20. Rxd1 Bxh2+ 21. Kxh2 Qxd3 22.
Rxd3 Red8 23. Nd7 Kg7 24. f3 h6 25. Rd6 f5 26. Nec5
Re8 27. Bd2 Re2 28. Kh1 Kf7 29. b3 Ke7 30. Nxb7 Rc8
31. Ndc5 Kf7 32. Kg1 Kg8 33. Rxg6+ Kf7 34. Rd6 Ne7
35. Rd7 Rg8 36. Nd6+ Kg6 37. Nc4 Kh5 38. Ne3 f4 39.
Ne6 fxe3 40. Nf4+ Kg5 41. Nxe2 exd2 42. Kf2 Kf6 43.
Nc3 Ke6 44. Rd4 Rc8 45. Ne4 d1=N+ 46. Rxd1 Rc2+ 47.
Kg1 Nf5 48. Rd2 Rxd2 49. Nxd2 Ke5 50. Kh2 Kf4 51.
b4 Ke3 52. Ne4 Nd4 53. a4 Nc2 54. b5 Nd4 55. Nd6
Ne6 56. a5 Kd3 57. g4 Kd4 58. b6 axb6 59. axb6 Nc5
60. Kg3 Kd5 61. b7 Na6 62. Nf7 Kc6 63. Nxh6 Kxb7
64. f4 Kc7 65. Kh4 Kd7 66. Kh5 Nc5 67. f5 Kd6 68.
f6 Ne6 69. Nf7+ Kc6 70. Kh6 Kd5 71. g5 Nc5 72. Kg6
Ne6 73. Kh5 Kd4 74. g6 Kc4 75. Ne5+ Kd5 76. Ng4 Ke4
77. Kh6 Kf5 78. g7 Nxg7 79. Kxg7 Kxg4 80. Kf7 Kf5
81. Ke7 Kg5 82. f7 Kf4 83. Kf6 Kf3 84. Ke5 Kg3 85.
Ke4 Kf2 86. Kd3 Kg1 87. Ke3 Kh2 88. f8=Q Kg3 89.
Qg8+ Kh3 90. Kf4 Kh2 91. Kf3 Kh3 92. Qg3# 1-0



Game D — SF 2500 vs. SOLIS ($S=5, W=3$), SOLIS–Black. Result: 1-0 SOLIS chooses a piece sacrifice whose short-term activity fails to compensate for the resulting long-term material deficit.

1. d4 d5 2. e3 Nd7 3. c4 e5 4. dxe5 dxc4 5. Nf3
Qe7 6. Bxc4 Nxe5 7. Nxe5 Qxe5 8. Qb3 Qh5 9. e4
Nf6 10. O-O Bd6 11. Qb5+ c6 12. Qxh5 Nxh5 13. Nc3
Be5 14. f4 Bc7 15. Be3 g6 16. Bb3 Be6 17. Rad1 Ng7
18. Bc2 f5 19. Bc5 Kf7 20. exf5 Nxf5 21. Ne4 b6
22. Ng5+ Kf6 23. Bxf5 Bd5 24. Rxd5 cxd5 25. Bd4+
Kxf5 26. h3 Bxf4 27. Bxh8 Kxg5 28. Bg7 Rg8 29. Bd4
Re8 30. Rf3 Re1+ 31. Kf2 Re6 32. g3 Bd2 33. Rf7
Re4 34. Bf6+ Kh6 35. g4 a6 36. Kg3 d4 37. h4 Be1+
38. Kh3 Re3+ 39. Kg2 Bxh4 40. Bxh4 a5 41. Bf6 a4
42. Rd7 Rd3 43. Rd5 b5 44. Rxb5 a3 45. Re5 Rd2+
46. Kf3 axb2 47. g5+ Kh5 48. Rb5 Rd3+ 49. Kf4 h6
50. gxh6+ Kxh6 51. Rxb2 g5+ 52. Kg4 Rd1 53. Bxg5+
Kg6 54. Rb6+ Kg7 55. Kf5 Kf7 56. Rb7+ Ke8 57. a4
Ra1 58. Re7+ Kf8 59. Rh7 Rxa4 60. Bh6+ Kg8 61. Re7
Rc4 62. Kg5 Rc5+ 63. Kg4 Ra5 64. Rg7+ Kh8 65. Re7
d3 66. Rd7 d2 67. Bxd2 Rc5 68. Bf4 Kg8 69. Be3 Rb5
70. Bh6 Re5 71. Kf4 Rc5 72. Ke4 Rb5 73. Kf4 Rb4+
74. Ke5 Rb1 75. Kf6 Rb2 76. Kg6 Rb6+ 77. Kh5 Rb2
78. Bg5 Rb6 79. Bh6 Re6 80. Kg5 Ra6 81. Rg7+ Kh8
82. Re7 Ra5+ 83. Kg6 Ra6+ 84. Kh5 Kg8 85. Rb7 Rc6
86. Rd7 Rc5+ 87. Bg5 Rc2 88. Kg6 Rc6+ 89. Bf6 Re6
90. Rd8+ Re8 91. Rxe8# 1-0

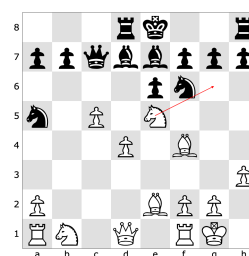


Game E — SOLIS ($S=5, W=3$) vs. Stockfish 2500, SOLIS–White. Result: 1-0 SOLIS sacrifices the exchange for long-term compensation and converts the ensuing advantage into a win.

A.5 EQUAL-DEPTH MATCHES: SOLIS D vs. STOCKFISH D

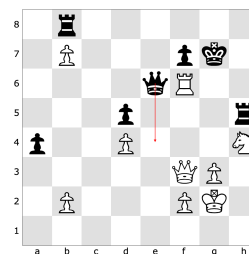
We include several illustrative games where SOLIS and Stockfish search to the same depth D . These examples emphasize how SOLIS' latent planning competes head-to-head without a rating cap on Stockfish. The left portion of each table encodes the moves of the game in Portable Game Notation (PGN) format; the right portion of each table displays a critical moment in the game.

1. d4 d5 2. e3 c5 3. Nf3 Nf6 4. Be2 e6 5. O-O Nc6
6. b3 cxd4 7. exd4 Be7 8. c3 Qc7 9. h3 Bd7 10. c4
dxc4 11. bxc4 Na5 12. Ne5 Rd8 13. Bf4 Bd6 14. c5
Be7 15. Ng6 Qc8 16. Nxh8 Bc6 17. Nc3 Kf8 18. Rc1
Qa8 19. Bc7 b5 20. Bxa5 Bxg2 21. d5 Bxh3 22. Qd3
Bf5 23. Qg3 b4 24. Rfd1 Bxc5 25. Qc7 Be7 26. Nb5
Nxd5 27. Rxd5 Qxd5 28. Bxb4 Qd7 29. Qxd7 Re8 30.
Rc8 g5 31. Qxe7+ Kg7 32. Qxf7+ Kh6 33. Bf8+ Rxf8 34.
Qxf8# 1-0



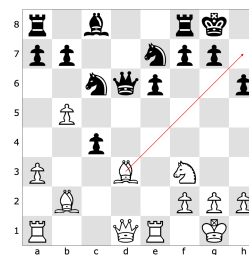
Game F — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 SOLIS sacrifices a knight in the middlegame and obtains a concrete tactical advantage.

1. d4 d5 2. e3 c5 3. Nf3 e6 4. Be2 cxd4 5. exd4
Bd6 6. O-O Nf6 7. Bg5 Nbd7 8. Nbd2 h6 9. Bh4 O-O
10. Re1 a5 11. Bf1 Bf4 12. c3 Qb6 13. Qc2 Qc7 14.
a4 b6 15. Bb5 Ba6 16. Bxa6 Rxa6 17. Qd3 Ra7 18. Re2
Qc8 19. Reel Nh5 20. Nf1 Nh6 21. N1d2 Nh5 22. Nf1
Bc7 23. Qd1 Nh6 24. Qd3 Nh5 25. Qd1 Nh6 26. Qd3
Rb7 27. Qb5 Ra7 28. N1d2 Nh5 29. h3 g5 30. Rac1
gxh4 31. c4 Nf4 32. Rc3 Qd8 33. g3 Nxh3+ 34. Kg2
Ng5 35. cxd5 h3+ 36. Kg1 h2+ 37. Kxh2 exd5 38. Rc6
Nf6 39. Nxg5 hxg5 40. Kg2 Re8 41. Rh1 Bd6 42. Nf3
Kg7 43. Nxg5 Rc7 44. Rxd6 Qxd6 45. Qf1 b5 46. axb5
Rcc8 47. Rh4 Rh8 48. Nf3 Ne4 49. Rf4 Rh5 50. Qe2
Qe6 51. Nh4 Rch8 52. b6 Nf6 53. Qf3 a4 54. b7 Rb8
55. Rxf6 Qe4 56. Rxf7+ Kh8 57. Qxe4 dxe4 58. Rc7
Rg5 59. Rc8+ Rg8 60. Rc7 Rg5 61. Rc8+ Rg8 62. Ng6+
Kg7 63. Ne7 Rf8 64. Rc7 Rf7 65. d5 Rbf8 66. Nf5+
Kg6 67. Rxf7 Kxf7 68. Nd6+ Kg6 69. Nc8 Kg5 70. b8=Q
Kg4 71. Qc7 Rf5 72. Nd6 e3 73. Qg7+ Rg5 74. f3+ Kh5
75. Qh7# 1-0



Game G — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 Stockfish accumulates an advantage before a late-middlegame blunder; SOLIS recognizes the opportunity and converts the resulting endgame.

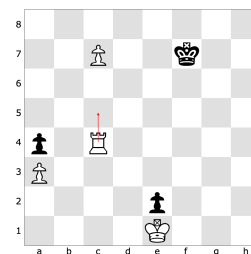
1. d4 c6 2. e4 d5 3. Nc3 Nf6 4. e5 Ng8 5. Bd3 e6
6. Nf3 c5 7. dxc5 Bxc5 8. O-O Nc6 9. a3 h6 10. b4
Bb6 11. Nb5 Bc7 12. Re1 Bb8 13. c4 Nge7 14. Nd6+
Bxd6 15. exd6 Qxd6 16. Bb2 O-O 17. b5 dxc4 18. Bh7+
Kh8 19. Qxd6 f6 20. Be4 c3 21. Bxc3 e5 22. bxc6 Bf5
23. Qxe7 Bxe4 24. Rxe4 bxc6 25. Nxe5 Rfe8 26. Nf7+
Kg8 27. Nxh6+ Kh7 28. Rae1 Rad8 29. Ba5 Rd4 30. g4
Rxe4 31. Rxe4 Rb8 32. Qxa7 Rd8 33. Rf4 Rd3 34. g5
fxg5 35. Rd4 Rxa3 36. Kf1 Rb3 37. Rb4 Rc3 38. h4
gxh4 39. Rd4 h3 40. Qe7 h2 41. Qe4+ Kxh6 42. Rd6+
Kg5 43. Rg6+ Kh5 44. Qg4# 1-0



Game H — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1-0 SOLIS sacrifices a bishop to create a discovered attack on Stockfish's queen and converts the resulting advantage.

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1. d4 d5 2. e3 c6 3. c4 e6 4. Nf3 Nf6 5. Be2 c5
6. O-O Nc6 7. b3 a6 8. Bb2 Be7 9. a3 cxd4 10. exd4
O-O 11. c5 b6 12. b4 bxc5 13. dxc5 Bb7 14. Nbd2 a5
15. b5 Bxc5 16. Rc1 Qb6 17. bxc6 Ng4 18. Rxc5 Ba6
19. Rc2 Nxf2 20. Rxf2 Bxe2 21. Qxe2 a4 22. c7 Rac8
23. Bd4 Qb5 24. Qxb5 Rfe8 25. Qb8 h6 26. Ne5 g5 27.
h4 Rf8 28. Ne4 dxe4 29. Rxf7 Rxb8 30. Rxf8+ Rxf8 31.
hxcg5 Rc8 32. gxh6 e3 33. Bxe3 Kh7 34. Rc4 Kg8 35.
Kh2 Kh7 36. g4 Rg8 37. g5 Ra8 38. Rg4 Kh8 39. Ba7
Rf8 40. Bb8 Rf2+ 41. Kg1 Rc2 42. Rc4 Rb2 43. Ba7
Rb1+ 44. Kg2 Kh7 45. Bf2 Rb2 46. g6+ Kxh6 47. Kh2
Rxf2+ 48. Kg3 Rg2+ 49. Kxg2 Kg7 50. Rc1 Kf6 51. Kh2
Kg7 52. Nd7 e5 53. Nf6 Kxf6 54. Rg1 Kg7 55. Rf1 Kh6
56. Rg1 Kg7 57. Rf1 Kxg6 58. Rc1 Kg7 59. Rc4 Kf7
60. Kg2 Ke7 61. Kf2 Kd6 62. Ke1 Kd5 63. Rc1 Ke4 64.
Ke2 Kd4 65. Rc2 e4 66. Rc1 Kd5 67. Ke1 Ke5 68. Rc4
Kd5 69. Rc2 Kd6 70. Rc1 Ke7 71. Rc4 Kf7 72. Rc5 e3
73. Kf1 Ke7 74. Rc4 Kf7 75. Kg2 e2 76. Kf2 Ke7 77.
Ke1 Kf7 78. Rc5 Kg7 79. Rc2 Kf6 80. Rc4 Kf7 81. Rc5
Kg6 82. Kd2 Kf7 83. Ke1 Ke7 84. Rc4 Kf7 85. Rc5 Kg6
86. Rc4 Kf7 87. Rc5 1/2-1/2



Game I — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS–White. Result: 1/2-1/2 SOLIS fails to prioritize the most forcing continuation in a position where many lines are already winning, resulting in a draw.

A.6 SOLIS VS. STOCKFISH RESULTS

For reproducibility of the Elo estimates, we report the complete SOLIS–Stockfish match outcomes for SOLIS search depths $S \in \{2, 3, 4, 5\}$. Each table indexes Stockfish Elo caps by row and SOLIS search width W by column. Entries give win–loss–draw counts for the corresponding matchup, with total match points in parentheses. Separate tables are provided for Base and Mini, and for unanchored and anchored scoring.

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 6: Complete Base-model match outcomes against Stockfish under unanchored planning, grouped by SOLIS search depth. Entries report win–loss–draw counts with match score in parentheses out of 100.

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1020
1021
1022
1023
1024
1025

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	32–32–36 (48)	27–45–28 (50)	–	–	1350	–	–	–	–
1400	30–41–29 (50)	33–27–40 (46)	–	–	1400	–	–	–	–
1450	29–44–27 (51)	33–40–27 (53)	–	–	1450	–	–	–	–
1500	31–38–31 (50)	31–39–30 (50)	–	–	1500	29–38–33 (48)	28–43–29 (50)	–	–
1550	38–35–27 (56)	23–50–27 (48)	–	–	1550	41–36–23 (59)	36–37–27 (54)	–	–
1600	34–30–36 (49)	29–37–34 (48)	–	–	1600	37–30–33 (52)	37–30–33 (52)	–	–
1650	24–45–31 (46)	28–36–36 (46)	–	–	1650	32–38–30 (51)	30–40–30 (50)	–	–
1700	–	–	–	–	1700	28–37–35 (46)	21–43–36 (42)	–	–
1750	–	–	–	–	1750	36–33–31 (52)	28–40–32 (48)	–	–
1800	–	–	37–31–32 (52)	37–36–27 (55)	1800	39–24–37 (51)	29–35–36 (46)	–	–
1850	–	–	34–24–42 (46)	34–31–35 (50)	1850	–	–	–	–
1900	–	–	40–28–32 (54)	40–29–31 (54)	1900	–	–	–	–
1950	–	–	39–29–32 (54)	39–29–32 (54)	1950	–	–	–	–
2000	–	–	38–34–28 (55)	42–28–30 (56)	2000	–	–	34–35–31 (52)	38–21–41 (48)
2050	–	–	41–27–32 (54)	40–26–34 (53)	2050	–	–	38–34–28 (55)	34–33–33 (50)
2100	–	–	35–30–35 (50)	36–26–38 (49)	2100	–	–	33–34–33 (50)	34–29–37 (48)
2150	–	–	–	–	2150	–	–	31–35–34 (48)	38–25–37 (50)
2200	–	–	–	–	2200	–	–	38–18–44 (47)	31–26–43 (44)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	45–26–29 (58)
2300	–	–	–	–	2300	–	–	40–22–38 (51)	34–26–40 (47)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	36–42–22 (57)	35–31–34 (50)	–	–	1600	–	–	–	–
1650	29–40–31 (49)	36–33–31 (52)	–	–	1650	–	–	–	–
1700	33–34–33 (50)	36–32–32 (52)	–	–	1700	39–30–31 (54)	34–34–32 (51)	–	–
1750	43–26–31 (56)	28–38–34 (47)	–	–	1750	37–29–34 (52)	32–30–38 (47)	–	–
1800	38–24–38 (50)	30–37–33 (48)	–	–	1800	42–20–38 (52)	33–31–36 (48)	–	–
1850	49–23–28 (60)	29–33–38 (46)	–	–	1850	36–24–40 (48)	43–28–29 (57)	–	–
1900	–	–	–	–	1900	45–26–29 (58)	46–31–23 (62)	–	–
1950	–	–	–	–	1950	40–22–38 (51)	37–35–28 (54)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	31–31–38 (46)	44–21–35 (54)	2150	–	–	–	–
2200	–	–	35–29–36 (50)	48–27–25 (62)	2200	–	–	27–37–36 (46)	33–38–29 (52)
2250	–	–	37–31–32 (52)	37–28–35 (51)	2250	–	–	30–33–37 (46)	38–31–31 (54)
2300	–	–	27–37–36 (46)	36–34–30 (53)	2300	–	–	38–37–25 (56)	33–39–28 (52)
2350	–	–	22–42–36 (43)	37–35–28 (54)	2350	–	–	32–29–39 (46)	29–36–35 (47)
2400	–	–	36–29–35 (50)	37–26–37 (50)	2400	–	–	30–26–44 (43)	30–32–38 (46)
2500	–	–	–	–	2500	–	–	26–34–40 (43)	43–25–32 (56)

Figure 7: Complete Mini-model match outcomes against Stockfish under unanchored planning, grouped by SOLIS search depth. Entries report win–loss–draw counts with match score in parentheses out of 100.

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Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32-46-22 (55)	41-47-12 (64)	-	-	1500	-	-	-	-
1550	39-38-23 (58)	37-50-13 (62)	-	-	1550	-	-	-	-
1600	32-44-24 (54)	33-51-16 (58)	-	-	1600	-	-	-	-
1650	33-41-26 (54)	29-46-25 (52)	-	-	1650	-	-	-	-
1700	34-38-28 (53)	22-53-25 (48)	-	-	1700	-	-	-	-
1750	37-35-28 (54)	30-49-21 (54)	-	-	1750	-	-	-	-
1800	29-42-29 (50)	36-42-22 (57)	-	-	1800	39-29-32 (54)	37-49-14 (62)	-	-
1850	-	-	-	-	1850	35-32-33 (51)	30-50-20 (55)	-	-
1900	-	-	-	-	1900	37-30-33 (52)	40-41-19 (60)	-	-
1950	-	-	-	-	1950	31-28-41 (45)	37-44-19 (59)	-	-
2000	-	-	34-40-26 (54)	43-31-26 (58)	2000	41-27-32 (54)	33-45-22 (56)	-	-
2050	-	-	46-38-16 (65)	36-38-26 (55)	2050	39-27-34 (52)	31-41-28 (52)	-	-
2100	-	-	34-45-21 (56)	34-31-35 (50)	2100	41-27-32 (54)	36-38-26 (55)	-	-
2150	-	-	46-31-23 (62)	30-43-27 (52)	2150	-	-	-	-
2200	-	-	31-35-34 (48)	35-28-37 (49)	2200	-	-	33-42-25 (54)	30-41-29 (50)
2250	-	-	34-34-32 (51)	40-29-31 (54)	2250	-	-	35-41-24 (56)	33-43-24 (54)
2300	-	-	32-32-36 (48)	37-34-29 (54)	2300	-	-	35-33-32 (52)	36-34-30 (53)
2350	-	-	-	-	2350	-	-	37-37-26 (56)	29-37-34 (48)
2400	-	-	-	-	2400	-	-	34-36-30 (52)	37-38-25 (56)
2450	-	-	-	-	2450	-	-	37-28-35 (51)	43-29-28 (58)
2500	-	-	-	-	2500	-	-	41-30-29 (56)	34-30-36 (49)
2550	-	-	-	-	2550	-	-	-	-
2600	-	-	-	-	2600	-	-	-	-
2650	-	-	-	-	2650	-	-	-	-
2700	-	-	-	-	2700	-	-	-	-
Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	-	-	-	-	1500	-	-	-	-
1550	-	-	-	-	1550	-	-	-	-
1600	-	-	-	-	1600	-	-	-	-
1650	-	-	-	-	1650	-	-	-	-
1700	-	-	-	-	1700	-	-	-	-
1750	-	-	-	-	1750	-	-	-	-
1800	-	-	-	-	1800	-	-	-	-
1850	-	-	-	-	1850	-	-	-	-
1900	-	-	-	-	1900	-	-	-	-
1950	37-26-37 (50)	41-42-17 (62)	-	-	1950	36-34-30 (53)	38-41-21 (58)	-	-
2000	43-20-37 (53)	31-43-26 (52)	-	-	2000	31-28-41 (45)	27-52-21 (53)	-	-
2050	38-27-35 (52)	40-38-22 (59)	-	-	2050	41-24-35 (53)	36-37-27 (54)	-	-
2100	42-25-33 (54)	32-48-20 (56)	-	-	2100	43-23-34 (54)	36-39-25 (56)	-	-
2150	36-23-41 (48)	31-47-22 (54)	-	-	2150	42-23-35 (54)	27-45-28 (50)	-	-
2200	46-13-41 (52)	39-39-22 (58)	-	-	2200	42-21-37 (52)	23-51-26 (48)	-	-
2250	-	-	-	-	2250	-	-	-	-
2300	-	-	-	-	2300	-	-	-	-
2350	-	-	-	-	2350	-	-	-	-
2400	-	-	-	-	2400	-	-	-	-
2450	-	-	40-39-21 (60)	39-38-23 (58)	2450	-	-	25-46-29 (48)	43-30-27 (58)
2500	-	-	38-38-24 (57)	41-40-19 (61)	2500	-	-	40-36-24 (58)	34-44-22 (56)
2550	-	-	44-20-36 (54)	42-29-29 (56)	2550	-	-	40-31-29 (56)	38-38-24 (57)
2600	-	-	46-17-37 (54)	37-23-40 (48)	2600	-	-	46-25-29 (58)	33-37-30 (52)
2650	-	-	41-18-41 (50)	44-22-34 (55)	2650	-	-	47-17-36 (56)	30-35-35 (48)
2700	-	-	45-16-39 (53)	42-22-36 (53)	2700	-	-	38-25-37 (50)	45-21-34 (56)

Figure 8: Complete Base-model match outcomes against Stockfish under anchored planning, grouped by SOLIS search depth. Entries report win-loss-draw counts with match score in parentheses out of 100.

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1116
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1118
1119
1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
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1132
1133

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	36–32–32 (52)	39–30–31 (54)	–	–	1350	–	–	–	–
1400	23–49–28 (48)	30–32–38 (46)	–	–	1400	–	–	–	–
1450	33–37–30 (52)	24–47–29 (48)	–	–	1450	–	–	–	–
1500	34–38–28 (53)	25–40–35 (45)	–	–	1500	29–40–31 (49)	35–33–32 (52)	–	–
1550	28–38–34 (47)	40–39–21 (60)	–	–	1550	34–35–31 (52)	37–29–34 (52)	–	–
1600	23–47–30 (46)	28–42–30 (49)	–	–	1600	32–36–32 (50)	32–39–29 (52)	–	–
1650	35–29–36 (50)	38–37–25 (56)	–	–	1650	34–30–36 (49)	37–33–30 (54)	–	–
1700	–	–	–	–	1700	36–30–34 (51)	34–35–31 (52)	–	–
1750	–	–	–	–	1750	34–25–41 (46)	34–30–36 (49)	–	–
1800	–	–	35–37–28 (54)	40–28–32 (54)	1800	32–29–39 (46)	32–31–37 (48)	–	–
1850	–	–	32–32–36 (48)	44–29–27 (58)	1850	–	–	–	–
1900	–	–	37–30–33 (52)	45–21–34 (56)	1900	–	–	–	–
1950	–	–	43–30–27 (58)	36–29–35 (50)	1950	–	–	–	–
2000	–	–	35–31–34 (50)	43–27–30 (56)	2000	–	–	46–32–22 (62)	38–27–35 (52)
2050	–	–	37–29–34 (52)	45–27–28 (58)	2050	–	–	37–27–36 (50)	39–26–35 (52)
2100	–	–	36–27–37 (50)	42–34–24 (59)	2100	–	–	29–34–37 (46)	48–17–35 (56)
2150	–	–	–	–	2150	–	–	37–33–30 (54)	39–25–36 (52)
2200	–	–	–	–	2200	–	–	37–31–32 (52)	41–23–36 (52)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	44–20–36 (54)
2300	–	–	–	–	2300	–	–	30–43–27 (52)	38–26–36 (51)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	26–45–29 (48)	31–33–36 (48)	–	–	1600	–	–	–	–
1650	37–39–24 (56)	28–32–40 (44)	–	–	1650	–	–	–	–
1700	23–39–38 (42)	29–41–30 (50)	–	–	1700	37–36–27 (55)	34–29–37 (48)	–	–
1750	36–29–35 (50)	38–30–32 (53)	–	–	1750	42–23–35 (54)	37–33–30 (54)	–	–
1800	35–24–41 (47)	33–36–31 (51)	–	–	1800	36–28–36 (50)	31–35–34 (48)	–	–
1850	41–21–38 (52)	30–28–42 (44)	–	–	1850	38–21–41 (48)	33–32–35 (49)	–	–
1900	–	–	–	–	1900	37–27–36 (50)	37–37–26 (56)	–	–
1950	–	–	–	–	1950	43–18–39 (52)	38–24–38 (50)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	39–23–38 (50)	32–30–38 (47)	2150	–	–	–	–
2200	–	–	33–40–27 (53)	38–25–37 (50)	2200	–	–	30–40–30 (50)	32–34–34 (49)
2250	–	–	36–31–33 (52)	30–27–43 (44)	2250	–	–	29–43–28 (50)	49–26–25 (62)
2300	–	–	32–33–35 (48)	37–28–35 (51)	2300	–	–	31–30–39 (46)	30–35–35 (48)
2350	–	–	35–34–31 (52)	40–25–35 (52)	2350	–	–	26–32–42 (42)	40–34–26 (57)
2400	–	–	28–26–46 (41)	26–38–36 (45)	2400	–	–	39–32–29 (55)	31–32–37 (47)
2500	–	–	–	–	2500	–	–	30–29–41 (44)	28–41–31 (48)

Figure 9: Complete Mini-model match outcomes against Stockfish under anchored planning, grouped by SOLIS search depth. Entries report win–loss–draw counts with match score in parentheses out of 100.